

Technical training.
Product information.

F25 LCI/F26 Complete Vehicle



BMW Service

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General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left-hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as a result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the technical training of the BMW Group and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

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F25 LCI/F26 Complete Vehicle

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1. Introduction

The F25 will receive a model revision from start of production April 2014.

The range of BMW X models will be extended with the X4 (development code F26) starting in July 2014.

In the following Technical Reference Manual ST1411 F25 LCI/F26 Complete Vehicle, are described in one Technical Reference Manual because both vehicles have many similarities.

1.1. Models

The following models will be available from July 2014:

1.1.1. F25 LCI



F25 LCI

F25 Models

- BMW X3 xDrive28i
- BMW X3 xDrive35i
- BMW X3 xDrive28d
- BMW X3 sDrive28i

F25 LCI/F26 Complete Vehicle

1. Introduction

1.1.2. F26



F26

F26 Models

- BMW X4 xDrive28i
- BMW X4 xDrive35i

F25 LCI and F26 Side views

A comparison of the side view of the F25 LCI with the F26 reveals that both vehicles are very similar from the front bumper to the A-pillar.



Side view of the F25 LCI/F26

F25 LCI/F26 Complete Vehicle

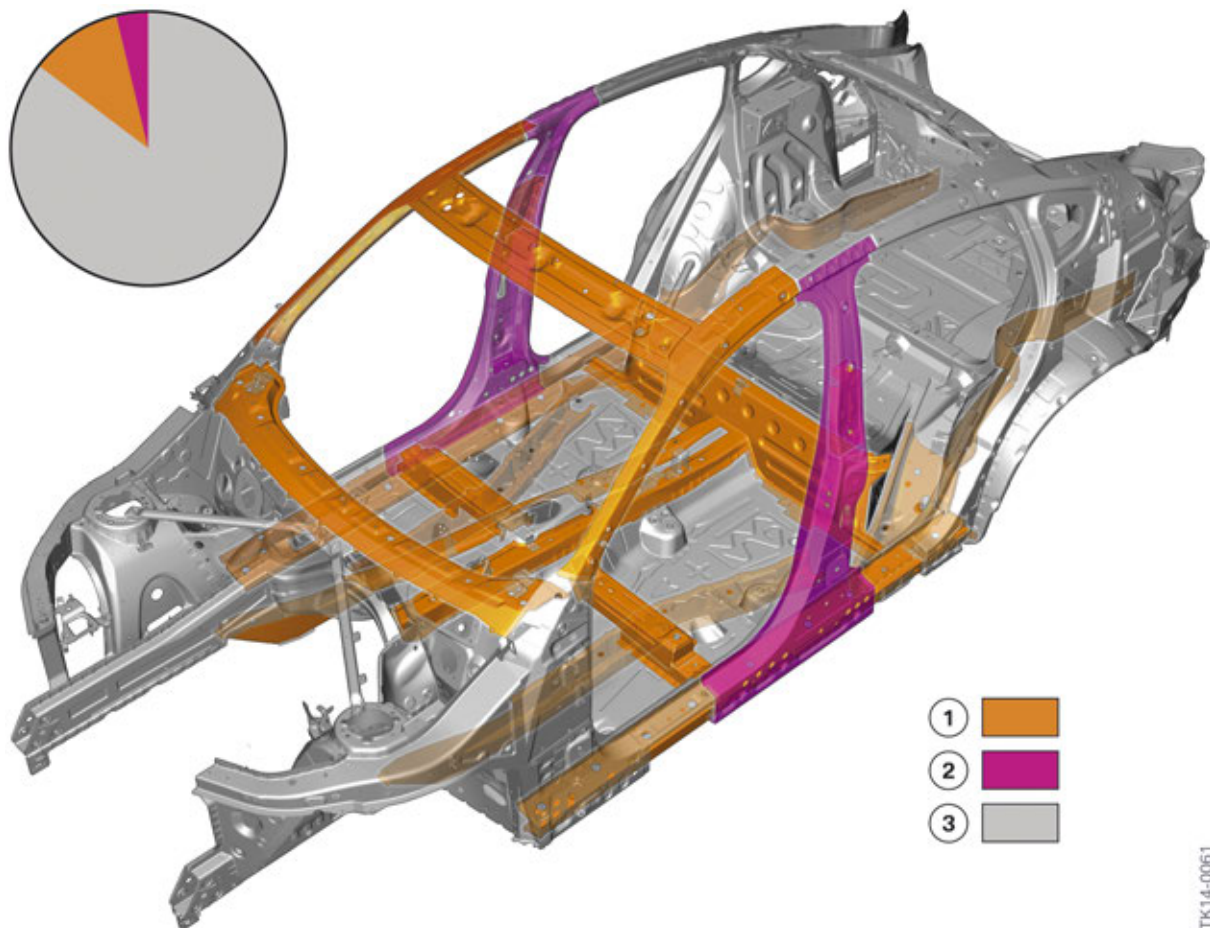
2. Body

The bodyshell of the F25 LCI/F26 is attached to the floor assembly of the F25.

2.1. Bodyshell F26

Here the material properties of the F26 are shown. The material properties of the F25 LCI have not changed in comparison to those of the F25.

Further information on the body of the F25 LCI can be found in the Technical Reference Manual ST1106 F25 Complete Vehicle.



F26 Material properties of bodyshell

Index	Explanation
1	Multi-phase steels (> 300 MPa), 8% of the body weight
2	Hot-formed steels (> 900 MPa), 2% of the body weight
3	Other steels (< 300 MPa), 90% of the body weight

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2. Body

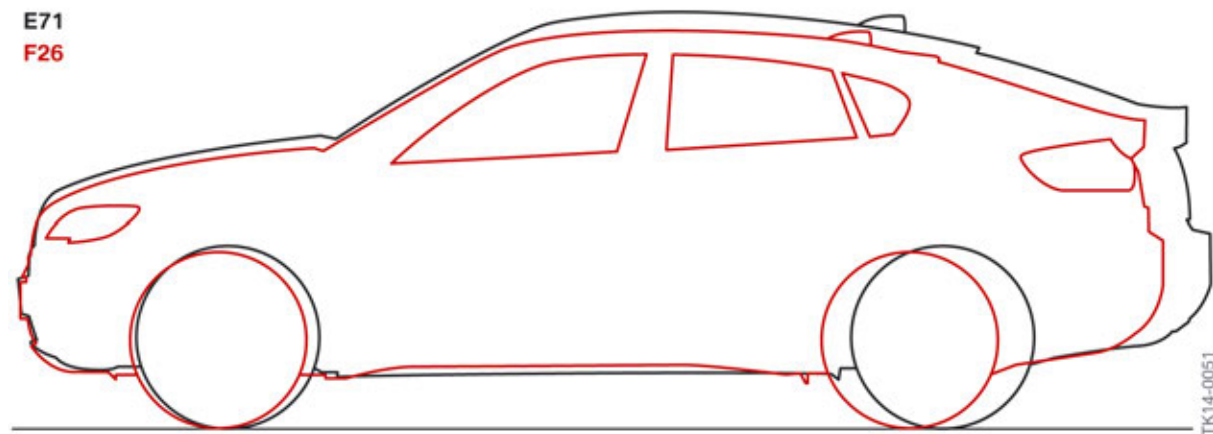
Multi-phase steels are steels where the structure consists of a number of phases. Higher-strength multi-phase steels with a yield strength of $R_{p0.2}$ of 300 to 600 MPa are, for example, dual-phase steels or TRIP steels. Super-strength multi-phase steels with a yield strength of $R_{p0.2}$ in excess of 600 MPa are, for example, complex-phase steels or martensitic-phase steels.

Hot-formed manganese-boron steels are super-strength steels with a yield strength of $R_{p0.2}$ in excess of 900 MPa.

To reduce the vehicle weight and still guarantee maximum stability of the bodyshell, high-strength and super-strength multi-phase steels are used, among other things. Super-strength hot-formed steels are also used.

2.2. Silhouette comparison of F26 with E71

The following graphic shows the F26 in a silhouette comparison with the E71.



F26 silhouette comparison with E71

	Unit	F26	E71	Difference
Vehicle height, empty	[mm]	1624	1679	-55
Front overhang	[mm]	876	858	+18
Wheelbase	[mm]	2810	2933	-123
Rear overhang	[mm]	1000	1086	-86
Vehicle length	[mm]	4680	4877	-197

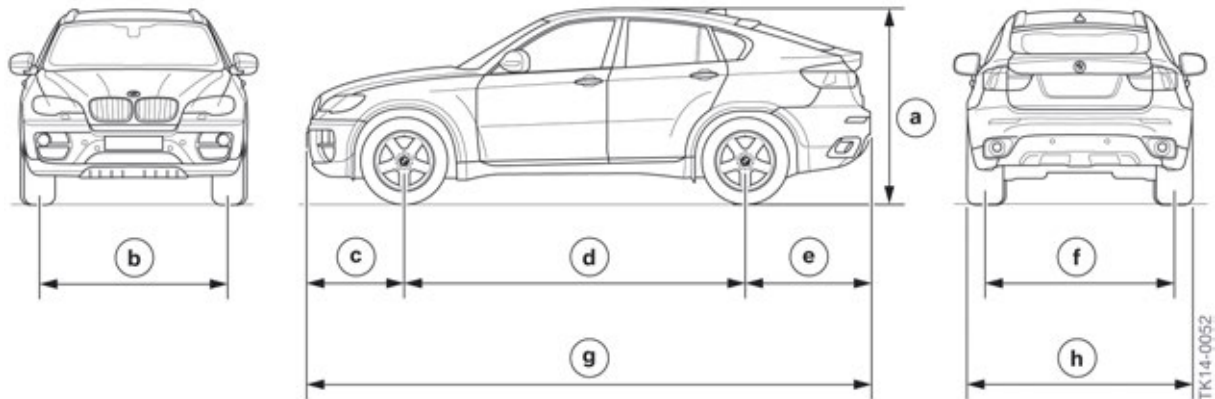
F25 LCI/F26 Complete Vehicle

2. Body

2.3. Outer dimensions

2.3.1. F26

Below you see the outer dimensions of the F26 in the European version.



F26 Outer dimensions

Index	Explanation	Unit	F26
a	Vehicle height, empty	[mm]	1624
b	Front track width, basic wheels	[mm]	1616
c	Front overhang	[mm]	876
d	Wheelbase	[mm]	2810
e	Rear overhang	[mm]	1000
f	Rear track width, basic wheels	[mm]	1632
g	Vehicle length	[mm]	4680
h	Width excl. exterior mirrors (vehicle width incl. exterior mirrors)	[mm]	1881 (2089)

F25 LCI/F26 Complete Vehicle

2. Body

2.3.2. Comparison of the dimensions of the F25 to the F25 LCI

	Unit	F25	F25 LCI	Change
Vehicle height, empty	[mm]	1678	1678	±0
Wheelbase	[mm]	2810	2810	±0
Vehicle length	[mm]	4648	4668	+20
Vehicle width excl. exterior mirrors	[mm]	1881	1881	±0
Vehicle width incl. exterior mirrors	[mm]	2098	2089	-9
Curb vehicle weight US	[kg] (lb)	1798 (4000)	1828 (4030)	+30

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2. Body

2.4. Interior equipment

2.4.1. Sun visor

The sun visors on the driver's and passenger's side can be moved approx. 10 cm and thus offer sun protection over a larger area of the driver's and passenger's window.



F25LCI/F26 variable sun visor

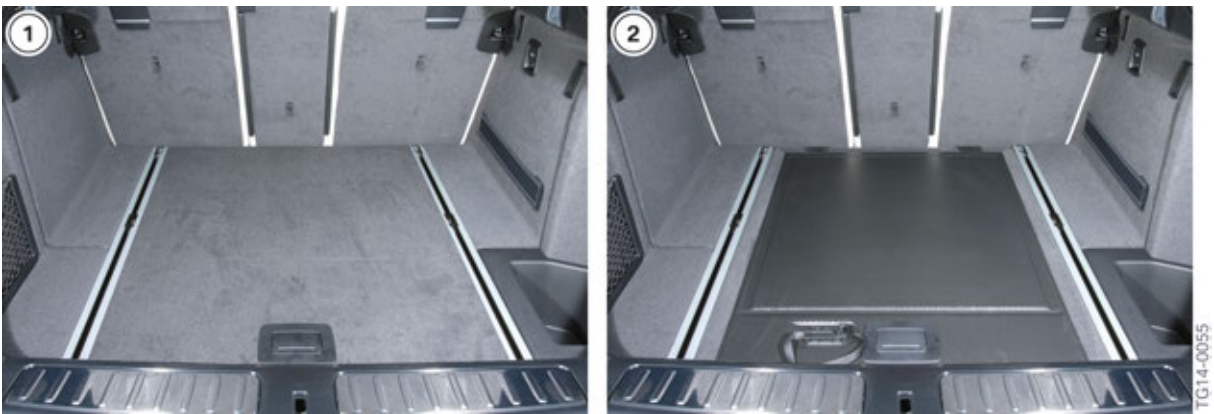
Index	Explanation
1	Sun visor, folded to the side
2	Sun visor, folded to the side with maximum extraction

F25 LCI/F26 Complete Vehicle

2. Body

2.4.2. Luggage compartment

Storage compartment package (SA 493) is standard in the F25 LCI/F26, it receives a reversible luggage compartment floor which has different materials on the top and bottom. By turning the luggage compartment floor over, the fabric covering or the anti-slip covering can be used.



F25 LCI/F26 luggage compartment

Index	Explanation
1	Luggage compartment floor with fabric covering
2	Luggage compartment floor with anti-slip covering

F25 LCI/F26 Complete Vehicle

2. Body

2.5. Exterior

The following are the changes to the exterior trim in comparison to the F25:

- Exterior mirror with integrated turn indicator.
- Modified headlight, optional with LED technology.
- More distinctive double radiator grille.
- New front and rear bumpers.



Differences compared to F25

Index	Explanation
1	F25 LCI
2	F25

The exterior mirror of the F25 LCI/F26 includes the turn indicator in LED technology. The width of the exterior mirror was also revised aerodynamically.

F25 LCI/F26 Complete Vehicle

2. Body

2.6. Acoustic measures

In order to improve the acoustics in the passenger compartment, in the F25 LCI/F26 insulating materials are installed at various places in the vehicle. For example, at the A-pillars foamed material is applied at the bottom in the outer area, which closes the gap between the body and the screwed-on side panel.



F25 LCI/F26 foamed material insert, A-pillar

Index	Explanation
1	Foamed material insert

F25 LCI/F26 Complete Vehicle

3. Powertrain

3.1. Overview of engines

In the following chapter you find an overview of the gasoline and diesel engines used in the F25 LCI/F26.

3.1.1. Gasoline engines

Model	Engine	Maximum power [kW (HP)]	Maximum torque [Nm (lb-ft)]	Displacement [cm ³]
BMW X4 xDrive28i BMW X3 xDrive28i BMW X3 sDrive28i	N20B20O0	180 (240) rpm	350 (260)	1997
BMW X4 xDrive35i BMW X3 xDrive35i	N55B30M0	225 (300)	407 (300)	2979

3.1.2. Diesel engine

Model	Engine	Maximum power [kW (HP)]	Maximum torque [Nm (lb-ft)]	Displacement [cm ³]
BMW X3 xDrive28d	N47D20O1	135 (180)	380 (280)	1995

Further information on the other engines used in the F25 LCI/F26 can be obtained in the following Technical Reference Manuals:

Engine	Technical Reference Manual
N20	ST1111 N20 Engine
N55	ST916 N55 Engine
N57	ST1305 N47TU Diesel Engine

F25 LCI/F26 Complete Vehicle

3. Powertrain

3.2. Automatic transmission

The 8HP transmission known from other BMW models is used as an automatic transmission.

In the F25 the sport automatic transmission is available as optional equipment (SA 2TB).
The following table provides an overview of the transmission variants used as standard and optional in the vehicle types:

Vehicle type	F25 LCI	F26
xDrive28i	Automatic transmission Optional Sports automatic transmission	Sports automatic transmission
xDrive35i	Automatic transmission Optional Sports automatic transmission	Sports automatic transmission
sDrive28i	Automatic transmission Optional Sports automatic transmission	—
xDrive28d	Automatic transmission	—

Information on the new features of the automatic transmission can be found in the Technical Reference Manual ST1309 5-Series LCI Complete Vehicle.

The torque converter with centrifugal pendulum is installed solely in conjunction with the N47 engines.

3.2.1. Sports automatic transmission

Additional functions are available for the optional equipment Sports automatic transmission (SA 2TB):

- Three different shift speeds, depending on the driving dynamics setting.
- Suppression of automatic gear shifts in manual mode.
- Launch control.

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4. Chassis and Suspension

In the F25 LCI/F26 the chassis and suspension known from the F25 is used. In this chapter the changes and new features are described.

4.1. Transfer box

In the F25 LCI/F26 the transfer box Active Torque Control 45L is used.

The following table provides an overview of the transfer box Active Torque Control 45L used for the first time in other BMW models:

Model	Used as of
E70	04/12
E71	04/12
F25	10/11
F15	11/13
F26	07/14

In this chapter only the changes to the transfer box Active Torque Control (ATC) 45L vis-a-vis the Active Torque Control (ATC) 450 are described.

The function of the transfer box Active Torque Control (ATC) 45L is identical to the function of the Active Torque Control (ATC) 450.

Further information on the transfer box Active Torque Control (ATC) 450 can be found in the Technical Reference Manual ST1106 F25 Complete Vehicle.

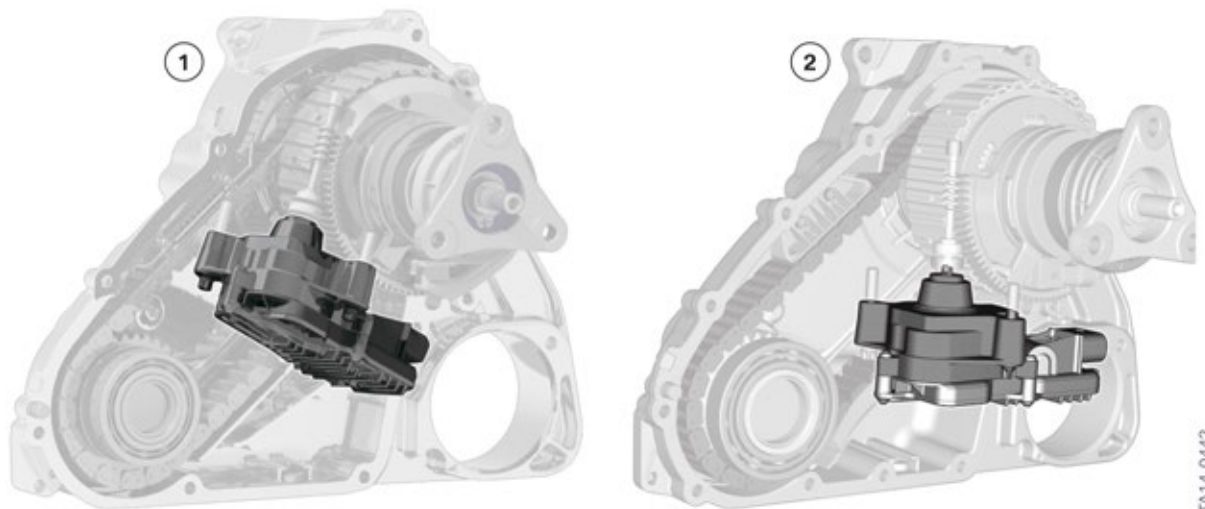
In the following table you see the key changes to the transfer box Active Torque Control (ATC) 45L:

Measures	Change
Optimized transmission housing	- Weight saving of approx. 1.5 kg - Modified chain drive
Adapted multidisc clutch	- Increased friction radius - Increased number of discs - Coil springs instead of disc springs - Discs are mounted directly on the shaft
Modified oil circuit	Optimization of the friction loss

F25 LCI/F26 Complete Vehicle

4. Chassis and Suspension

The transfer box Active Torque Control (ATC) 45L can be identified by its diagonal mounted actuator.



Different installation position of the control units in the transfer box

Index	Explanation
1	Active Torque Control 45L
2	Active Torque Control 450

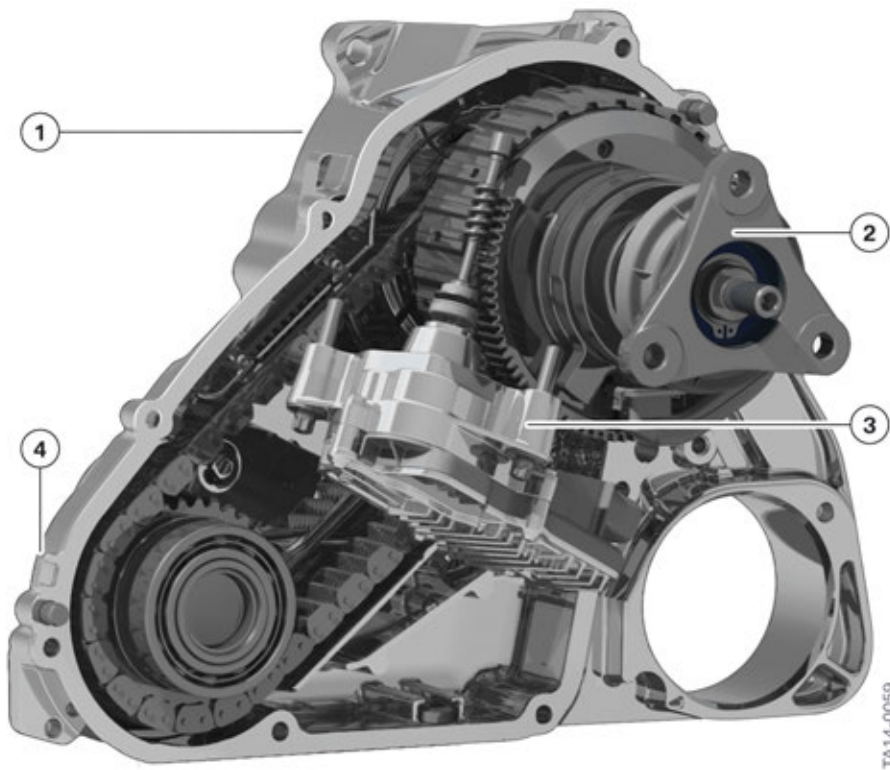
The transfer box Active Torque Control (ATC) 45L is designed for power transmission up to 1100 Nm.



During a replacement of the transfer box Active Torque Control (ATC) 45L, ensure that it is not replaced with the transfer box Active Torque Control (ATC) 450, as otherwise its servomotor rotates in the opposite direction. The same also applies for the replacement of the transfer box control unit.

F25 LCI/F26 Complete Vehicle

4. Chassis and Suspension



Structure of the transfer box Active Torque Control 45L

Index	Explanation
1	Connection to automatic transmission
2	Connection to the rear axle
3	Integrated control unit with servomotor
4	Connection to the front axle

F25 LCI/F26 Complete Vehicle

4. Chassis and Suspension

4.1.1. Mechanics

The discs in the transfer box Active Torque Control 45L are fitted directly on the input shaft and not on a clutch housing as with the transfer box Active Torque Control 450.



Multidisc clutch, transfer box Active Torque Control 45L

Index	Explanation
1	Worm drive for actuators
2	Multidisc clutch

F25 LCI/F26 Complete Vehicle

4. Chassis and Suspension

4.2. Displays, xDrive status

In the central information display (CID) the current xDrive status of the vehicle can be displayed with the following information:

- Compass direction.
- Slope of the vehicle in longitudinal direction with specification of percentage and degree.
- Inclination of the vehicle in transverse direction with specification of degree.

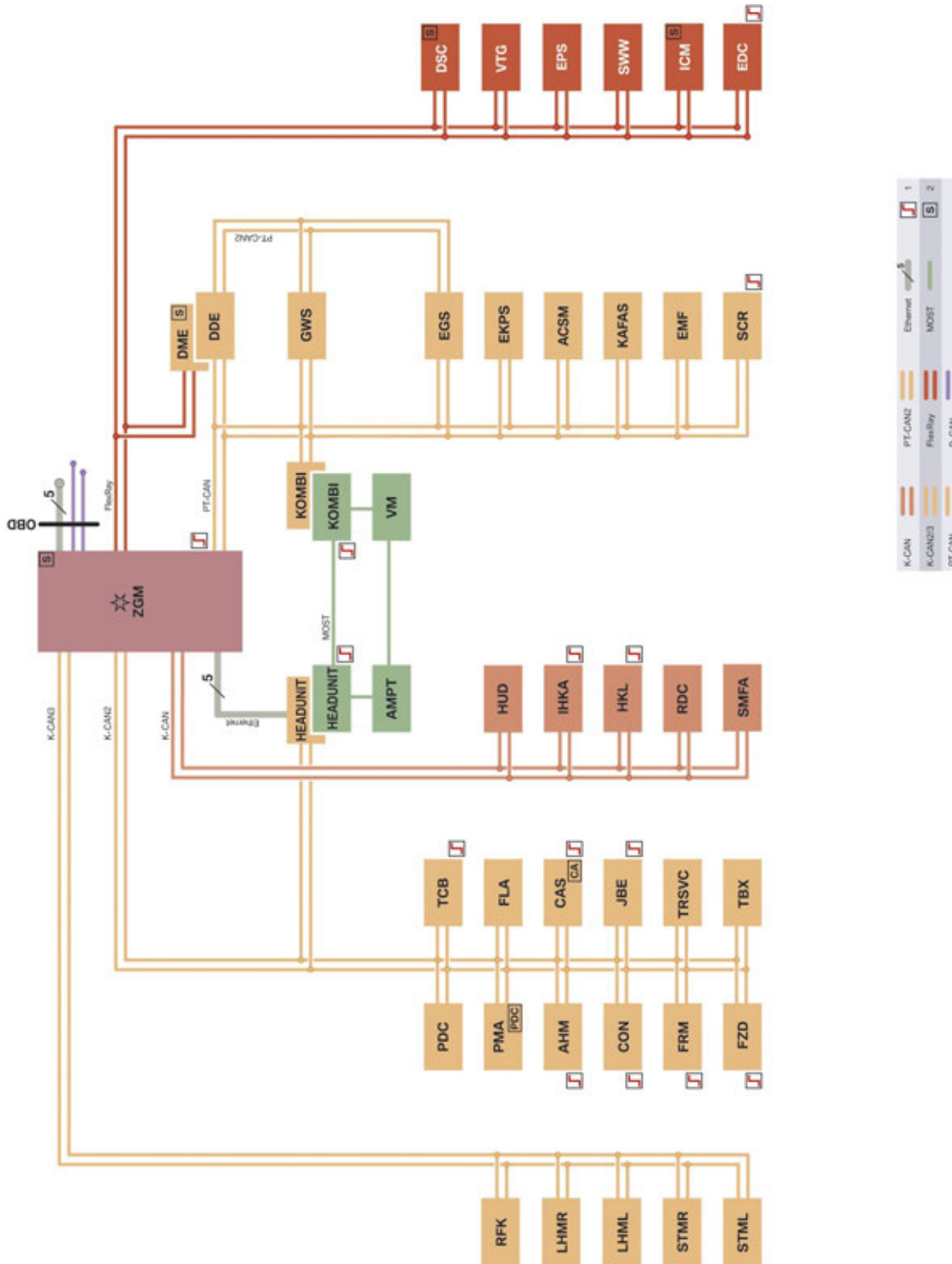


F25 LCI/F26 display of xDrive status in the central information display

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.1. Bus overview



TE14-0062

F25 LCI/F26 bus overview

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

Index	Explanation
1	Control units with wake-up authorization
2	Start-up node control units for starting and synchronizing the FlexRay bus system
ACSM	Crash Safety Module
AHM	Trailer module
AMPT	Amplifier Top (top high fidelity amplifier)
CAS	Car Access System
CON	Controller
DDE	Digital Diesel Electronics (F25 28d only)
DME	Digital Engine Electronics
DSC	Dynamic Stability Control
EDC	Electronic Damper Control
EGS	Electronic transmission control
EMF	Electromechanical parking brake
EKPS	Electronic fuel pump control
EPS	Electromechanical Power Steering
FLA	High-beam assistant
FRM	Footwell module
FZD	Roof function center
GWS	Gear selector switch
HEADUNIT	Headunit
HKL	Automatic operation of tailgate
HUD	Head-Up Display
ICM	Integrated Chassis Management
IHKA	Integrated automatic heating / air conditioning
JBE	Junction box electronics
KAFAS	Camera-based driver support systems
KOMBI	Instrument cluster
LHML	LED main light module on the left
LHMR	LED main light module on the right
PMA	Parking Manoeuvring Assistant
PDC	Park Distance Control
TPMS	Tire Pressure Monitoring System
SCR	Selective Catalytic Reduction (F25 28d only)
SMFA	Driver's seat module

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5. General Vehicle Electronics

Index	Explanation
STML	Left-hand headlight driver module
STMR	Right headlight driver module
SWW	Lane change warning
TCB	Telematic Communication Box
TBX	Touchbox
TR SVC	Top rear side view camera
VM	Video Module (Not for US)
VTG	Transfer box
ZGM	Central gateway module

5.2. Voltage supply

5.2.1. Components

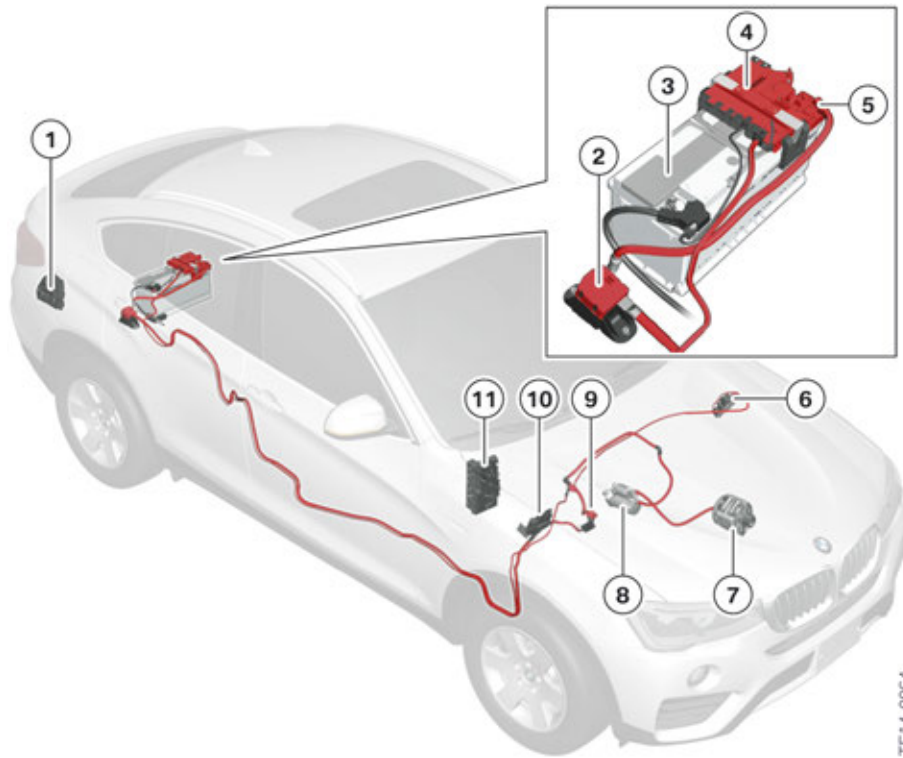
Five separate power distribution boxes with fuse blocks are installed in the F25LCI/F26:

- Power distribution box, engine compartment
- Power distribution box, (PDM) (only with gasoline engines)
- Junction box for the power distribution box
- Power distribution box, luggage compartment
- Battery power distribution box.

The junction box comprises the junction box power distribution box and junction box electronics (JBE).

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics



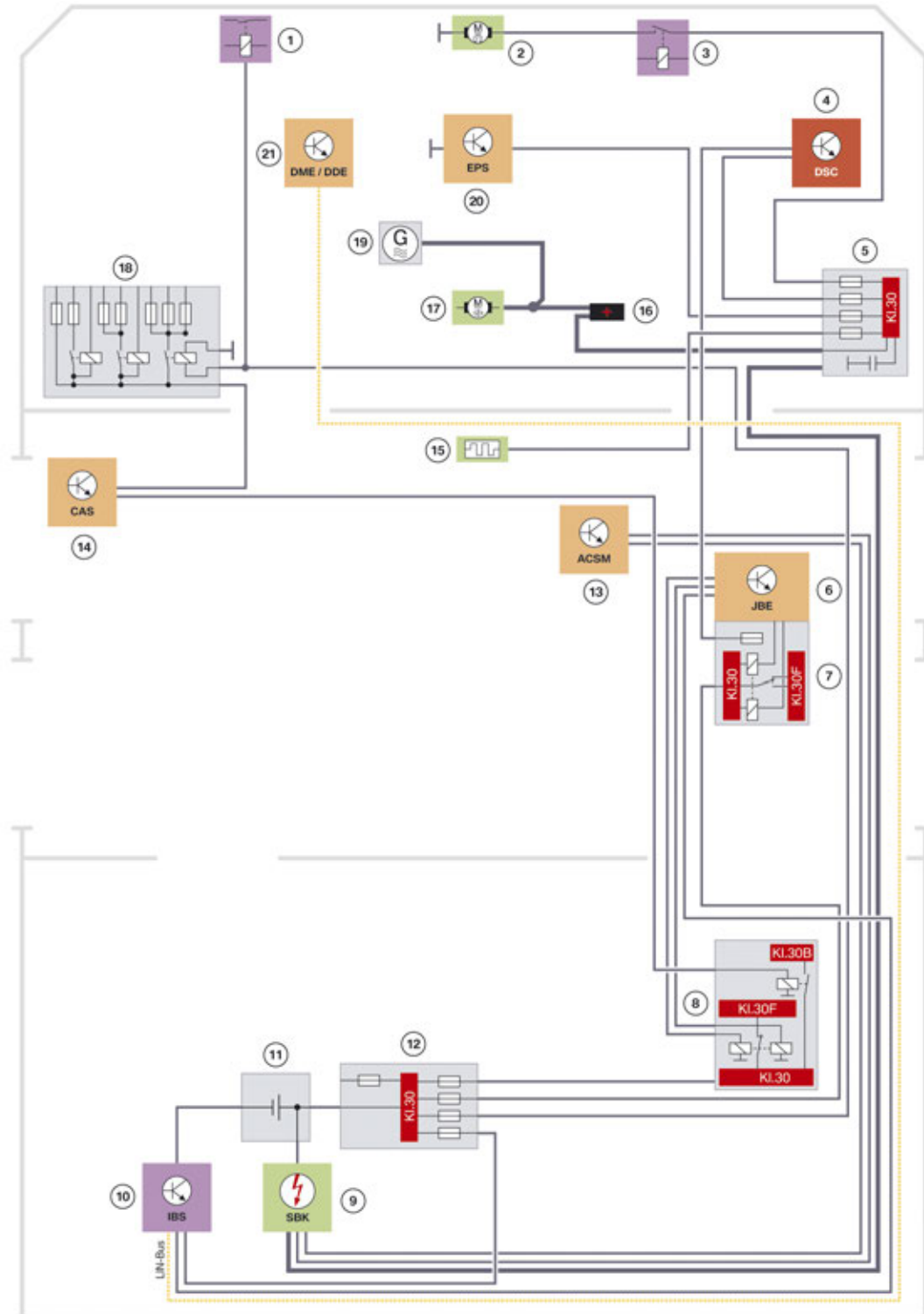
F25 LCI/F26 components of the voltage supply

Index	Explanation
1	Power distribution box, luggage compartment
2	Transfer point, luggage compartment
3	Battery
4	Battery power distribution box
5	Safety battery terminal (SBK)
6	Power Distribution Module (PDM)
7	Alternator
8	Starter motor
9	Positive battery connection point
10	Power distribution box, engine compartment
11	Power distribution box, junction box

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.2.2. System wiring diagram



TE14-0063

F25 LCI/F26 system wiring diagram for voltage supply

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

Index	Explanation
1	DDE main relay
2	Electric fan
3	Relay for electric fan
4	Dynamic Stability Control (DSC)
5	Power distribution box, engine compartment
6	Junction box electronics (JBE)
7	Junction box for the power distribution box
8	Power distribution box, luggage compartment
9	Safety battery terminal (SBK)
10	Intelligent Battery Sensor (IBS)
11	Battery
12	Battery power distribution box
13	Crash Safety Module (ACSM)
14	Car Access System (CAS)
15	Electric auxiliary heater (only for diesel fuel vehicles)
16	Positive battery connection point
17	Starter motor
18	Power distribution box, (PDM) (only with gasoline engines)
19	Alternator
20	Electronic Power Steering (electromechanical power steering) (EPS)
21	Digital Engine Electronics (DME) or Digital Diesel Electronics (DDE)

5.3. Exterior lights

The F25 LCI/F26 is equipped in the basic equipment with halogen headlights and fog lights.

5.3.1. Headlights

The following headlights and systems are offered as standard and optional equipment for the F25 LCI/F26:

- Headlight washer system (SA 502, country-specific) optional F25/F26
- Bi-xenon headlight (SA 522) standard on F26 and F25 xDrive35i
- Adaptive Headlight (SA 524) standard on F26 and F25 xDrive35i
- Adaptive LED headlight (SA 552) optional F25/F26
- LED fog light (SA 5A1) standard on F25/F26
- High-beam assistant (SA 5AC) optional F25/F26.

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

The cornering light is incorporated into the fog lights.

5.3.2. Bi-xenon headlights

The bi-xenon headlights is standard on F26 and F25 xDrive35i (SA 522).



F25 LCI/F26 bi-xenon headlight

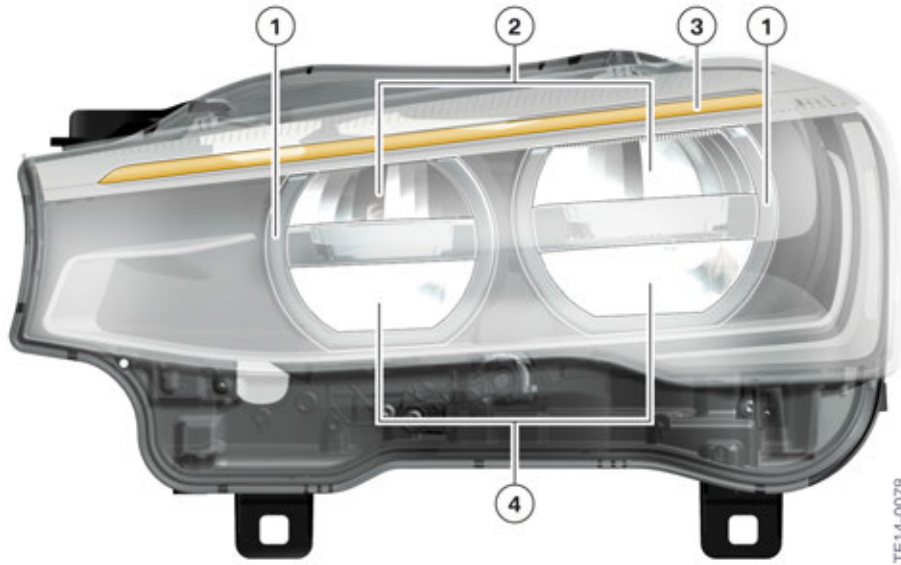
Index	Explanation
1	Side light/Daytime driving lights via lighting rings
2	Bi-xenon headlights
3	Turn indicator

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.3.3. Adaptive LED Headlights

The adaptive LED headlight is available in the F25 LCI/F26 as optional equipment (SA 552).



F25 LCI/F26 LED headlight

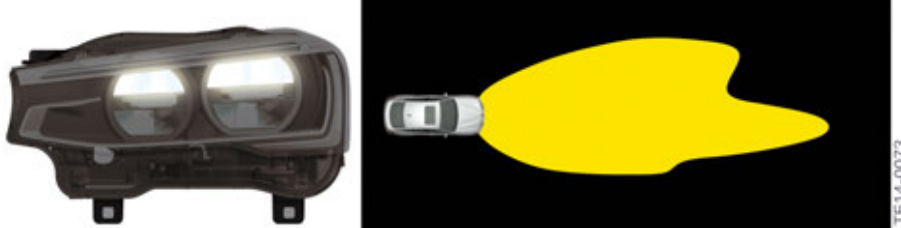
Index	Explanation
1	Side light/Daytime driving lights via lighting rings
2	Low-beam headlight
3	Turn indicator
4	High-beam headlight

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.3.4. Functions of the adaptive LED headlight

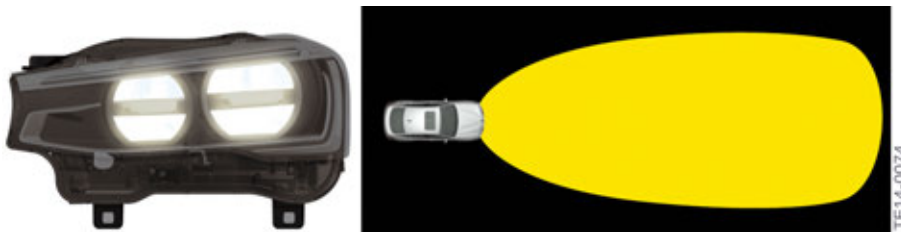
Low-beam headlight



F25 LCI/F26 low-beam headlight

With the low-beam headlight the two upper halves of the adaptive LED headlight are active.

High beam



F25 LCI/F26 high beam

All LEDs (low beam/high beam) are active when the high beam is switched on.

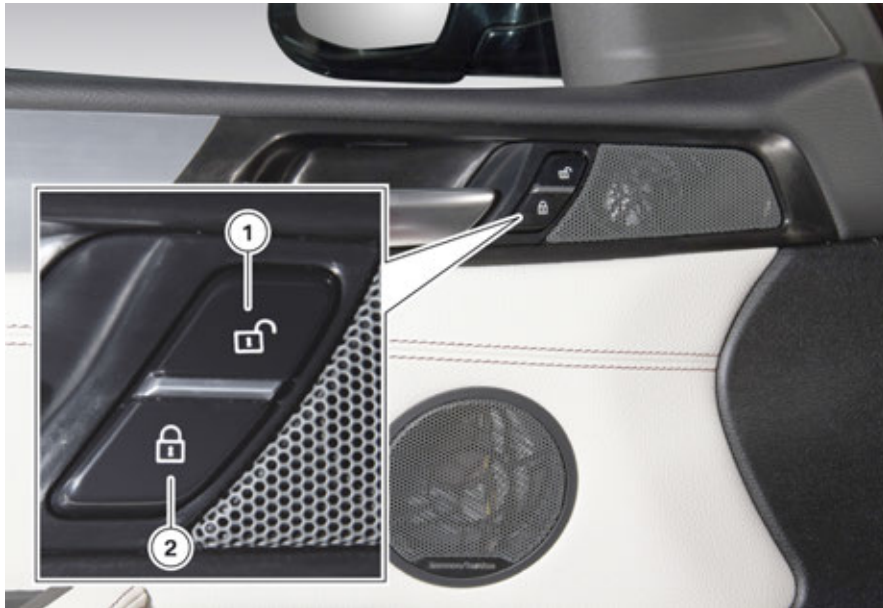
F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.4. Locking and security systems

5.4.1. Central locking system

The central locking button is designed as a rocker switch.



F25 LCI/F26 central locking button

Index	Explanation
1	Unlocking
2	Locking

F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.4.2. Tailgate release

The button for the tailgate release is installed in the switch block on the driver's side. The tailgate can be opened and closed using the button.



F25 LCI/F26 switch block on driver's door

Index	Explanation
1	Safety switch, rear power window
2	Power window, rear left
3	Power window, front left
4	Folding exterior mirror in and out
5	Exterior mirror adjustment
6	Tailgate release
7	Exterior mirror, left/right, activate/deactivate automatic folding function
8	Power window, front right
9	Power window, rear right

Non-contact tailgate opening

In conjunction with Comfort Access (SA 322) optional equipment, the F25 LCI/F26 has the function of the non-contact tailgate opening:

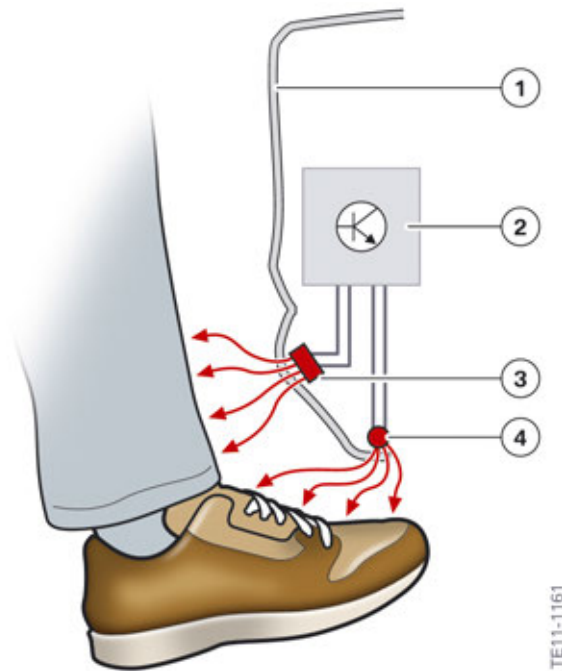
- Comfort Access (SA 322) optional
- Power tailgate (SA 316) standard equipment

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5. General Vehicle Electronics

The operation of the contact-free tailgate opening is effected by means of targeted foot movement to and from the bumper. Two sensors detect the movement without contact via capacitive measurement.

Further information on the non-contact tailgate opening can be found in the Technical Reference Manual ST1113 F30 Complete Vehicle.



F25 LCI/F26 non-contact tailgate opening by foot movement

Index	Explanation
1	Cross-section, rear bumper
2	Control unit for non-contact tailgate opening
3	Sensor at top for non-contact tailgate opening
4	Sensor at bottom for non-contact tailgate opening

The following components are involved in the function of the non-contact tailgate opening:

- Central gateway module
- Car Access System
- Footwell module
- Junction box electronics
- Tailgate lock
- Control unit for non-contact tailgate opening
- Comfort Access antenna, bumper, rear
- Upper sensor, non-contact tailgate opening (3)
- Lower sensor, non-contact tailgate opening (4)

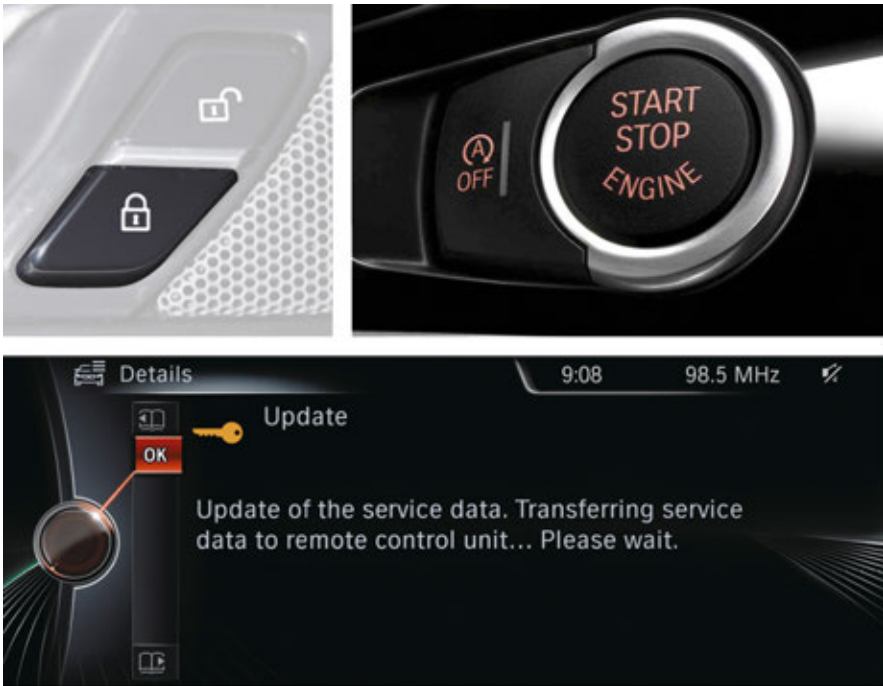
F25 LCI/F26 Complete Vehicle

5. General Vehicle Electronics

5.5. Updating Service key data

Proceed as follows to transfer current service data to the ID transmitter during Service:

- Keep the central locking button pressed down and then press the start/stop button (terminal 15 ON). Then release both buttons.
- Hold the ID transmitter within 10 seconds at the key symbol on the right side of the steering column shroud.



F25 LCI/F26 service data update

F25 LCI/F26 Complete Vehicle

6. Driver Assistance Systems

6.1. Special equipment

The driver assistance systems of the F25 LCI/F26 are already known from other vehicles. The following table provides an overview of the available driver assistance systems:

Assistance system	Technical reference manual for further information
Lane departure warning	ST1309 5-Series LCI Complete Vehicle
Speed limit information	ST1309 5-Series LCI Complete Vehicle
Parking Manoeuvring Assistant	ST1002 F10 Complete Vehicle
High-beam assistant	ST811 F01/F02 Complete Vehicle
Rear view camera	ST1312 F15 Complete Vehicle
Surround View	ST1312 F15 Complete Vehicle
Lane change warning	ST811 F01/F02 Complete Vehicle
Forward collision warning with city braking function	ST1309 5-Series LCI Complete Vehicle
Pedestrian warning with city braking function	ST1309 5-Series LCI Complete Vehicle
ACC Stop&Go	ST1309 5-Series LCI Complete Vehicle

With the introduction of the optional equipment Active Driving Assistant (SA 5AS) and ACC Stop&Go + Active Driving Assistant (SA 5AT) some of the assistance systems are available individually and some as part of the two new optional equipment offers. The following table provides an overview.

Optional equipment	Function
Driving Assistant (SA 5AS)	Lane departure warning Camera-based collision warning Forward collision warning with city braking function Pedestrian warning with city braking function
ACC Stop&Go + Active Driving Assistant (SA 5AT)	Lane departure warning Forward collision warning with braking function Pedestrian warning with city braking function ACC Stop&Go Distance information (only in conjunction with (SA 610) Head up display)
Speed limit information (SA 8TH)	Speed limit information
Park Distance Control (SA 508)	Park Distance Control (PDC)
Parking Manoeuvring Assistant (SA 5DP)	Parking Manoeuvring Assistant
High-beam assistant (SA 5AC)	High-beam assistant
Lane change warning (SA 5AG)	Lane change warning
Rear view camera (SA 3AG)	Rear view camera
Top & side view cameras (SA 5DL)	Top & side view cameras

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6. Driver Assistance Systems

6.2. Sensors

The following table provides an overview of the sensors used in the assistance systems. Depending on the equipment the same system can be realized using different sensors.

Assistance system	KAFAS* camera	Radar sensor
Lane departure warning	X	
Lane change warning		X
Road sign recognition	X	
ACC Stop&Go	X	X
Camera-based collision warning	X	
Front collision warning with city braking function	X	
Pedestrian warning with city braking function	X	
High-beam assistant	X	

*KAFAS = camera-based driver support systems

F25 LCI/F26 Complete Vehicle

7. Display and Controls

7.1. Intelligent Safety button

The Intelligent Safety button is installed with the following optional equipment:

- Active Driving Assistant (SA 5AS)
- ACC Stop & Go + Active Driving Assistant (SA 5AT).



F25 LCI/F26 Intelligent Safety button

Index	Explanation
1	Intelligent Safety button

7.1.1. Displays

Briefly pressing the Intelligent Safety button causes a change in status of the assistance systems. The assistance systems can be activated or deactivated separately or individualized. Pressing the Intelligent Safety button for an extended period completely activates or deactivates the systems.



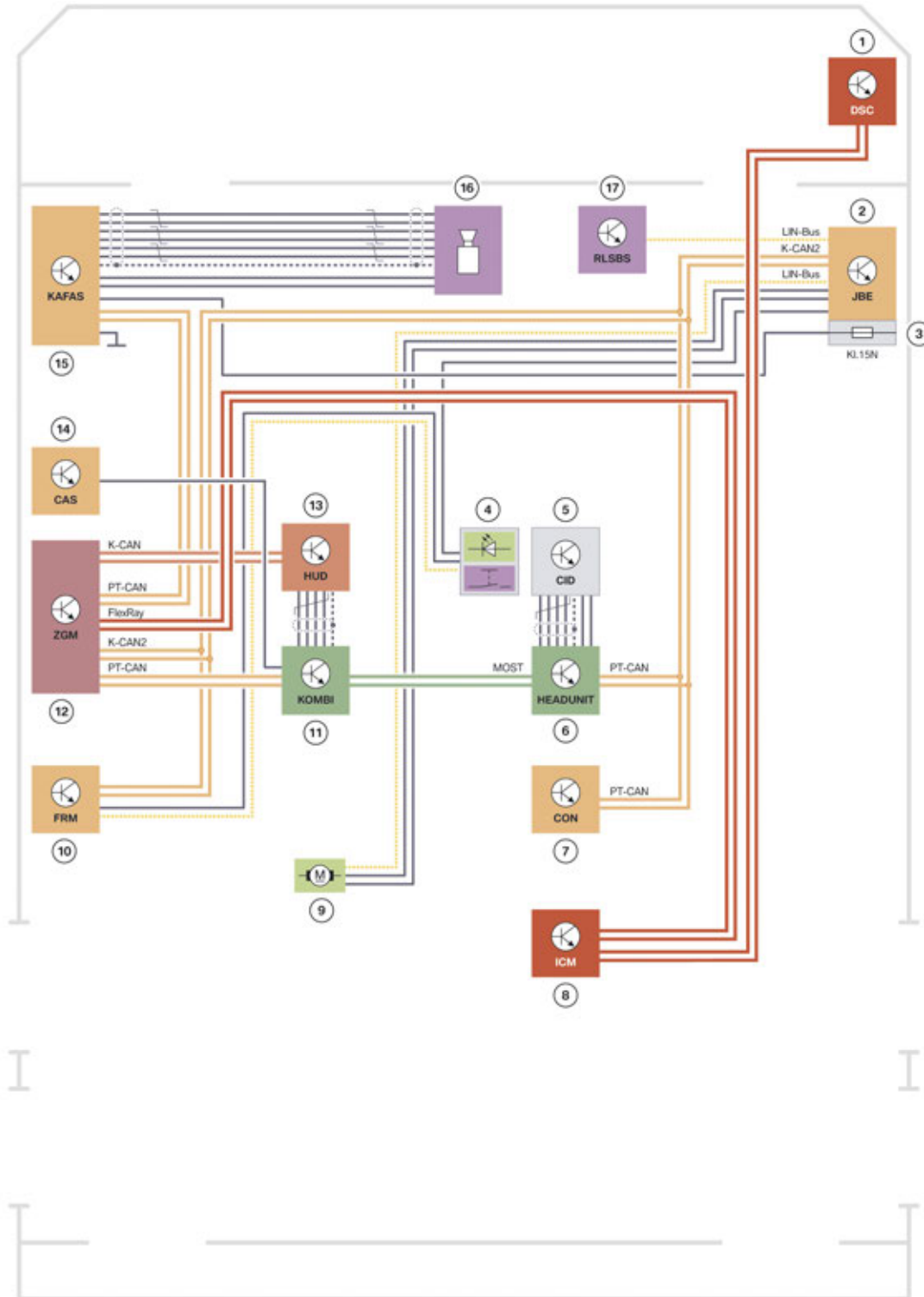
F25 LCI/F26 display of LED Intelligent Safety button

Index	Explanation
A	All assistance systems activated
B	Assistance systems can be individually adjusted
C	All assistance systems are deactivated

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7. Display and Controls

7.1.2. System wiring diagram



TE14-0058

F25 LCI/F26 system wiring diagram for camera-based driver support systems

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7. Display and Controls

Index	Explanation
1	Dynamic Stability Control (DSC)
2	Junction Box Electronics (JBE)
3	Fuse in the power distribution box, junction box
4	Intelligent Safety button
5	Central Information Display (CID)
6	Headunit
7	Controller (CON)
8	Integrated Chassis Management (ICM)
9	Vibration actuator in the steering wheel; connect at the steering column switch cluster (SZL)
10	Footwell module (FRM)
11	Instrument cluster (KOMBI)
12	Central Gateway Module (ZGM)
13	Head-Up Display (HUD)
14	Car Access System (CAS)
15	Camera-based driver support systems (KAFAS)

F25 LCI/F26 Complete Vehicle

7. Display and Controls

7.2. Instrument cluster

7.2.1. Variants

Three different instrument clusters can be installed depending on optional equipment.



F25 LCI/F26 instrument clusters

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7. Display and Controls

Index	Instrument cluster	Display	Equipment
1	Base	2.7" TFT display; resolution 320 x 120 pixels	Basic equipment
2	Dependent on optional equipment	5.7" TFT display; resolution 640 x 160 pixels	• Active Driving Assistant (SA 5AS) • ACC Stop & Go + Active Driving Assistant (SA 5AT) • Professional navigation system (SA 609).
3	High version	9.2" TFT display; resolution 960 x 160 pixels	Head-up Display (SA 610)

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7. Display and Controls

7.3. Head-Up Display

The Head-Up Display (HUD) (SA 610) contains various functions aimed at enhancing road safety and ride comfort. This includes the following displays:

- Vehicle speed
- Set speed for Cruise control with braking function
- Active cruise control with Stop & Go function
- Navigation system
- Check Control messages
- Speed limit information
- Lane departure warning
- Proactive driving assistant
- Audio lists

Head-Up Display

7.3.1. Demo mode

There is a Demo mode for the Head-Up Display (HUD) with which the various displays can be shown.

The Service menu must be activated to activate the Demo mode.

Activate Service menu:

- Call up main menu
- Move controller **to the front for longer than 10 seconds**
- **Turn controller to the right 3 steps**
- **Turn controller to the left 3 steps**
- **Turn controller to the right 1 step**

F25 LCI/F26 Complete Vehicle

7. Display and Controls

- **Turn controller to the left 1 step**
- **Turn controller to the right 1 step**
- Press controller **once**

The Service menu is then added as an additional submenu in the "Settings" menu.

Activate Demo mode:

The Demo mode is visible in the "Settings" menu of the HUD when the Service menu is activated.

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7. Display and Controls



Demo mode Head-Up Display

Index	Explanation
1	Service menu
2	"Settings" menu
3	Demo mode Head-Up Display in the "Head-Up Display" menu

7.4. Controller

The following variants of the controller are used for the F25 LCI/F26:

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7. Display and Controls

- Controller with five buttons in vehicles with Professional radio (already included in the basic equipment)
- Controller with seven buttons **with** touchpad in vehicles with Navigation Professional (SA 609).

7.4.1. Controller with touchpad

A controller with touchpad is installed in conjunction with the optional equipment Navigation Professional (SA 609). Further information on this controller is available in the Technical Reference Manual "ST1309 5-Series LCI Complete Vehicle".

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8. Information and Communication

8.1. Headunits

The following table gives you an overview of the headunits available for the F25 LCI/F26:

Optional equipment	Headunit	CID	Controller	Navigation
Radio BMW Professional (standard equipment)	Headunit Basic	6.5"	5-button	No
Professional navigation (SA 609)	Headunit High	8.8"	7-buttons with touchpad	Yes

8.1.1. Headunit High

The Headunit High known from other BMW models is also installed in the F25 LCI/F26 as optional equipment Navigation Professional (SA 609).

Further information on the Headunit High can be found in the Technical Reference Manual ST1211 Headunit High

8.1.2. Headunit Basic

The F25 LCI/F26 receives the new Headunit Basic.

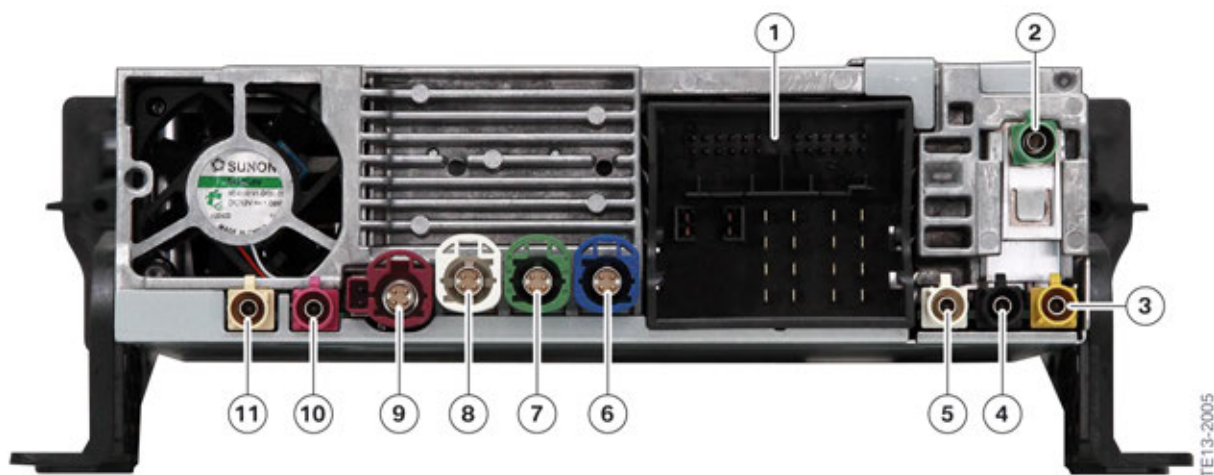
The Headunit Basic is installed with:

- Radio Professional (installed as standard)

Radio Professional

The F25 LCI/F26 receives a Professional radio as standard.

The Professional radio receives 6.5" display and a controller with 5 buttons.



Rear view, Professional radio

F25 LCI/F26 Complete Vehicle

8. Information and Communication

Index	Explanation
1	Main connector
2	DAB L band aerial (Not for US) Satellite radio for US market
3	DAB band 3 aerial (Not for US) Satellite for radio US market
4	AM/FM1 aerial
5	FM2 aerial
6	USB2 connection; connection for customer smartphone via the telephone base plate
7	USB3 connection; connection for Telematic Communication Box (TCB)
8	USB1 connection; customer access at the Aux-IN USB port at the Center Stack (also for data import/export)
9	Automotive Pixel Link connection and voltage supply of the central information display (CID)
10	Preparation of WLAN aerial
11	Connection for Bluetooth aerial

F25 LCI/F26 Complete Vehicle

8. Information and Communication

8.2. Telephone systems

Two telephone systems are available for the F25 LCI/F26:

- Hands-free system with USB interface (SA 6NH) standard equipment.
- Convenient telephone system with extended smartphone connection (SA 6NS) optional equipment.

The hands-free system with USB interface is an element of the optional equipment Navigation Professional (SA 609).

The following table provides an overview of the different telephone systems.

	SA 6NH	SA 6NS
Maximum number of connected mobile phones	4	4
Maximum number of connected mobile phones which can be used at the same time	1	2*
USB connection in centre console	X	X
Bluetooth hands-free mode	X	X
Bluetooth audio streaming	—	X
Base plate	—	X
Voice processing system	—	X
Display of phone book	X	X
Office functions	—	X
External aerial connection	—	X
Number of microphones	1	2

* Only one main telephone and one additional telephone can be connected via Bluetooth at the same time.



Bayerische Motorenwerke Aktiengesellschaft
Qualifizierung und Training
Röntgenstraße 7
85716 Unterschleißheim, Germany